AERO40006 - Mathematics 1

Tutorial 2

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Question 1

Evaluate these integrals using a change of variables (including the option of trigonometric substitutions),

(1)	$\int \sin(7x+1)dx$	(2)	$\int \cot(x)dx$
(3)	$\int \frac{3x^2-6x}{x^3-3x^2+1}dx$	(4)	$\int \frac{1}{\sqrt{1-x^2}}dx$

(1) Let $u = 7x+1$, $\frac{du}{dx} = 7$ $\therefore dx = \frac{1}{7}du$

$\therefore \int \sin(7x+1)\,dx = \int \frac{\sin u}{7} du = -\frac{1}{7}\cos u + C = -\frac{1}{7}\cos(7x+1) + C$

(2) $\int \cot(x)\,dx = \int \frac{1}{\tan(x)} dx = \int \frac{\cos(x)}{\sin(x)} dx$

Let $u = \sin(x)$ $\therefore \frac{du}{dx} = \cos(x)$ $\therefore dx = \frac{du}{\cos(x)}$

$\therefore \int \cot(x)\,dx = \int \frac{1}{\sin(x)} du = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin(x)| + C$

(3) Let $u = x^3 - 3x^2 + 1$ $\therefore \frac{du}{dx} = 3x^2 - 6x$ $\therefore dx = \frac{du}{3x^2-6x}$

$\therefore \int \frac{3x^2-6x}{x^3-3x^2+1} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|x^3-3x^2+1| + C$

(4) Let $x = \sin\theta$ $\therefore \frac{dx}{d\theta} = \cos\theta$ $\therefore dx = \cos\theta\, d\theta$

$\therefore \int \frac{\cos\theta}{\sqrt{1-\sin^2\theta}} d\theta = \int 1\, d\theta = \theta + C = \arcsin x + C$

Question 2

Evaluate these integrals using integration by parts,

(1)	$\int 2xe^x dx$	(2)	$\int x\sin(x)dx$
(3)	$\int x^2\ln(x)dx$	(4)	$\int x\cos(3x)dx$

(1)

	D	I
+	$2x$	e^x
−	2	e^x
+	0	e^x

$$\therefore \int 2xe^x dx = 2xe^x - 2e^x + C$$

(2)

	D	I
+	x	$\sin(x)$
−	1	$-\cos(x)$
+	0	$-\sin(x)$

$$\therefore \int x\sin(x)\,dx = -x\cos(x) + \sin(x) + C$$

(3)

	D	I
+	$\ln(x)$	x^2
−	$\frac{1}{x}$	$\frac{1}{3}x^3$
+	$-\frac{1}{x^2}$	$\frac{1}{12}x^4$
−	...	

$$\therefore \int x^2\ln(x)\,dx = \frac{1}{3}x^3\ln(x) - \int \frac{1}{3}x^2\,dx$$

$$= \frac{1}{3}x^3\ln(x) - \frac{1}{9}x^3 + C$$

(4)

	D	I
+	x	$\cos(3x)$
−	1	$\frac{1}{3}\sin(3x)$
+	0	$-\frac{1}{9}\cos(3x)$

$$\therefore \int x\cos(3x)\,dx = \frac{1}{3}x\sin(3x) + \frac{1}{9}\cos(3x) + C$$

Question 3

Use partial fractions, evaluate the following integrals,

(1)	$\int \frac{3x+4}{(x+1)(x+2)}dx$	(2)	$\int \frac{x+1}{(x-1)^2}dx$	(3)	$\int \frac{5x^2}{(x^2+1)(2x-1)}dx$

(1) $\int \frac{3x+4}{(x+1)(x+2)} dx = \int \frac{1}{x+1} dx + \int \frac{2}{x+2} dx$

$= \ln|x+1| + 2\ln|x+2| + C$

(2) $\int \frac{x+1}{(x-1)^2} dx = \int \frac{2}{(x-1)^2} dx + \int \frac{1}{x-1} dx$

$= \frac{-2}{(x-1)} + \ln|x-1| + C$

(3) $\int \frac{5x^2}{(x^2+1)(2x-1)} dx = \int \frac{5}{2x-1} dx + \int \frac{-5}{2x^3-x^2+2x-1} dx$

$= \frac{5}{2}\ln|2x-1| - \frac{5}{6x^2-2x+2} \ln|2x^3-x^2+2x-1| + C$

Question 4

Evaluate the following definite integrals:

(1)	$\int_0^1 (3x+4)^{-3} dx$	(2)	$\int_1^2 x\left(1+x^2\right)^{1/2} dx$
(3)	$\int_0^\pi \cos(4x) dx$	(4)	$\int_0^{\pi/2} \sin^2(x)\cos(x) dx$

(1) $\int_0^1 (3x+4)^{-3} dx = \left[-\frac{1}{6}(3x+4)^{-2}\right]_0^1 = \left(-\frac{1}{294}\right) - \left(-\frac{1}{96}\right) = \frac{11}{1568}$

(2) $u = 1+x^2 \quad \therefore \frac{du}{dx} = 2x \quad \therefore dx = \frac{du}{2x}$

$\int_1^2 x(1+x^2)^{\frac{1}{2}} dx = \int_1^5 \frac{1}{2}u^{\frac{1}{2}} du = \frac{1}{2}\left[\frac{2}{3}u^{\frac{3}{2}}\right]_1^5 = 2.78$

(3) $\int_0^\pi \cos(4x) dx = \left[\frac{1}{4}\sin(4x)\right]_0^\pi = 0 - 0 = 0$

(4) $u = \sin(x) \quad \therefore \frac{du}{dx} = \cos(x) \quad \therefore dx = \frac{du}{\cos(x)}$

$\int_0^{\frac{\pi}{2}} \sin^2(x)\cos(x) dx = \int_0^1 u^2 du = \left[\frac{1}{3}u^3\right]_0^1 = \frac{1}{3}$

Question 5

Evaluate the following integrals that involve recursive relations.

1. Let
$$I_m = \int_0^{\pi} \cos^m(x)dx, \ n = 0, 1, 2, \ \ldots$$
Derive a formula relating I_m and I_{m-2}(for $m \geq 2$). Use this formula to calculate I_4.

2. Given that
$$I_n = \int_0^{\pi} e^x \sin^n(x)dx, \ n = 0, 1, 2, \ \ldots$$
show that
$$(n^2+1)I_n = n(n-1)I_{n-2}.$$
Verify that
$$I_5 = \frac{3}{13}(e^{\pi}+1).$$

1. $I_m = \int_0^{\pi} \cos(x)\cos^{m-1}(x)\,dx = [\sin(x)\cos^{m-1}(x)]_0^{\pi} + (m-1)\int_0^{\pi} \sin^2(x)\cos^{m-2}(x)\,dx$

	D	I
+	$\cos^{m-1}(x)$	$\cos(x)$
−	$-(m-1)\sin(x)\cos^{m-2}(x)$	$\sin(x)$

$= [\sin(x)\cos^{m-1}(x)]_0^{\pi} + (m-1)\int_0^{\pi} (1-\cos^2(x))\cos^{m-2}(x)\,dx$

$= [\sin(x)\cos^{m-1}(x)]_0^{\pi} + (m-1)\int_0^{\pi} \cos^{m-2}x\,dx - (m-1)\int_0^{\pi}\cos^m(x)\,dx$

$= (0-0) + (m-1)I_{m-2} - (m-1)I_m$

$\therefore\ I_m = (m-1)I_{m-2} - (m-1)I_m$

$\therefore\ mI_m = (m-1)I_{m-2}$

$\therefore\ I_m = \frac{m-1}{m}I_{m-2}$

$\therefore\ I_4 = \frac{4-1}{4}I_2 = \frac{3}{4}\int_0^{\pi}\cos^2x\,dx = \frac{3}{4}\int_0^{\pi}\frac{1+\cos 2x}{2}\,dx$

$= \frac{3}{8}\int_0^{\pi}(1+\cos 2x)\,dx = \frac{3}{8}[x+\frac{1}{2}\sin 2x]_0^{\pi} = \frac{3}{8}(\pi-0) = \frac{3}{8}\pi$

Question 5

Evaluate the following integrals that involve recursive relations.

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$$I_m = \int_0^{\pi} \cos^m(x)dx, \; n \; = 0, 1, 2, \; \ldots$$
Derive a formula relating I_m and I_{m-2}(for $m \geq 2$). Use this formula to calculate I_4.

2. Given that
$$I_n = \int_0^{\pi} e^x \sin^n(x)dx, \; n \; = 0, 1, 2, \; \ldots$$
show that
$$(n^2+1)I_n = n(n \; - 1)I_{n-2}.$$
Verify that
$$I_5 = \frac{3}{13}(e^{\pi}+1).$$

2. $I_n = \int_0^{\pi} e^x \sin^n(x)\,dx = [e^x \sin^n(x)]_0^{\pi} - n\,[\cos(x)\sin^{n-1}(x)\,e^x]_0^{\pi}$
$+ \int e^x[-n\sin^n(x) + n(n-1)\cos^2(x)\sin^{n-2}(x)]dx$

D	I
$+ \sin^n(x)$	e^x
$- n\cos(x)\sin^{n-1}(x)$	e^x
$+ -n\sin^n(x) + n(n-1)\cos^2(x)\sin^{n-2}(x)$	e^x

$= 0 - 0 - n\int e^x \sin^n(x)\,dx$
$+ n(n-1)\int e^x(1-\sin^2(x))\sin^{n-2}(x)\,dx$
$= -n\int e^x \sin^n(x)\,dx + n(n-1)\int e^x \sin^{n-2}(x)\,dx$
$- n(n-1)\int e^x \sin^n(x)\,dx$
$= -n\,I_n + n(n-1)\,I_{n-2} - n(n-1)\,I_n$

$\therefore (1+n+n^2-n)\,I_n = n(n-1)\,I_{n-2}$

$\therefore (n^2+1)\,I_n = n(n-1)\,I_{n-2}$

$\therefore I_n = \frac{n(n-1)}{n^2+1} I_{n-2}$

$\therefore I_5 = \frac{20}{26} I_3 = \frac{10}{13}\left(\frac{6}{10} I_1\right) = \frac{6}{13}\int_0^{\pi} e^x \sin(x)\,dx$

D	I
$+ \sin(x)$	e^x
$- -\cos(x)$	e^x
$+ -\sin(x)$	e^x

$\because \int e^x \sin(x)dx = e^x\sin(x) + e^x\cos(x) - \int e^x \sin(x)\,dx$ $\quad\therefore \int e^x \sin(x)\,dx = \frac{1}{2}e^x\sin(x) + \frac{1}{2}e^x\cos(x)$

$\therefore I_5 = \frac{3}{13}[e^x \sin x + e^x\cos(x)]_0^{\pi}$

$= \frac{3}{13}(e^{\pi}+1)$

Question 6

Calculate the area delimited by the $x-$axis, a circle of radius $\sqrt{2}$ and the parabola $y = x^2$ in the first quadrant. See figure 1.

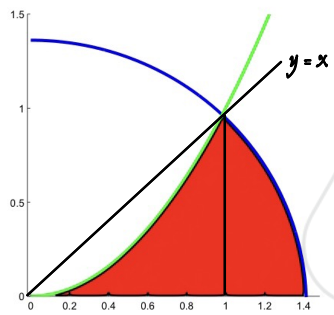


Figure 1: Area delimited by the graph of $y = x^2$, circle with radius $\sqrt{2}$ and $x-$axis.

$y = x^2 = \sqrt{2-x^2}$ $\quad \therefore x^4 = 2 - x^2$ $\quad \therefore x^4 + x^2 - 2 = 0$ $\quad \therefore (x^4+2)(x^2-1) = 0$

$\therefore x = 1$ or -1 $\quad \therefore x = 1$

$$\text{Area} = \frac{1}{8}\pi(\sqrt{2})^2 - \int_0^1 (x - x^2)\,dx$$

$$= \frac{1}{4}\pi - \left[\frac{1}{2}x^2 - \frac{1}{3}x^3\right]_0^1$$

$$= \frac{1}{4}\pi - \left(\frac{1}{2} - \frac{1}{3}\right)$$

$$= \frac{1}{4}\pi - \frac{1}{6}$$

Question 7

Calculate the volume of the solid of revolution generated by rotating the finite region bounded by the graphs of curves $y = \sqrt{x-1}$ and $y = (x-1)^2$ about the $y-$axis. See figure 2.

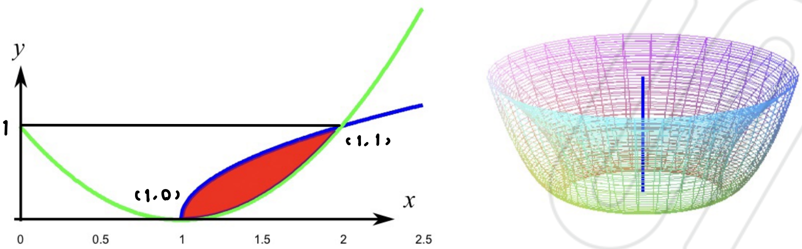


Figure 2: Area delimited by the curves $y = \sqrt{x-1}$ (blue) and $y = (x-1)^2$ (green) and the solid of revolution obtained after its rotation around the $y-$axis.

If $y = \sqrt{x-1} = (x-1)^2$

Let $a = x-1$ $\therefore a^{\frac{1}{2}} = a^2$ $\therefore a^2 - a^{\frac{1}{2}} = 0$ $\therefore a^{\frac{1}{2}}(a^{\frac{3}{2}} - 1) = 0$ $\therefore a^{\frac{1}{2}} = 0$ and $a^{\frac{3}{2}} = 1$

$\therefore x - 1 = 0$ or 1 $\therefore x = 1$ or 2 $\therefore y = 0$ or 1

$\therefore$ intercept points $(1, 0)$ and $(2, 1)$

$y = \sqrt{x-1} = (x-1)^{\frac{1}{2}}$ $\therefore x - 1 = y^2$ $\therefore x = 1 + y^2$

$y = (x-1)^2$ $\therefore y^{\frac{1}{2}} = x - 1$ $\therefore x = 1 + y^{\frac{1}{2}}$

$$\therefore V = \pi \int_0^1 [(1+y^{\frac{1}{2}})^2 - (1+y^2)^2] dy$$

$$= \pi \int_0^1 (1 + 2y^{\frac{1}{2}} + y - 1 - 2y^2 - y^4) dy$$

$$= \pi \int_0^1 (-y^4 - 2y^2 + y + 2y^{\frac{1}{2}}) dy$$

$$= \pi \left[-\frac{1}{5}y^5 - \frac{2}{3}y^3 + \frac{1}{2}y^2 + \frac{4}{3}y^{\frac{3}{2}} \right]_0^1$$

$$= \pi \left(\frac{29}{30} - 0 \right) = \frac{29}{30}\pi$$

Question 8

(i) Express the following complex numbers in the form $re^{i\theta}$:

$$(a)\ 5-12i \qquad (b)\ -1+4i \qquad (c)\ 2-i\ln 3$$

(ii) Express the following complex numbers in the form $a+ib$:

$$(a)\ \sqrt{3}e^{\frac{i\pi}{6}} \qquad (b)\ e^{\frac{7i\pi}{4}} \qquad (c)\ e^{-\frac{i\pi}{4}} \qquad (d)\ \mathrm{e}^{i}$$

(i) (a) $13e^{i5.107}$

(b) $\sqrt{17}e^{i1.816}$

(c) $2.282e^{i5.781}$

(ii) (a) $\frac{3}{4}+\frac{\sqrt{3}}{4}i$

(b) $\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i$

(c) $\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i$

(d) 1

Question 9

Prove that for any complex numbers $z, w \in \mathbb{C}$,

(i) $|z|^2 = z\bar{z}$

(ii) $|\text{Re}(z)| \leq |z|$

(iii) $|z + w| \leq |z| + |w|$

What is the geometrical interpretation of the inequality (iii)?

[Hint: for (iii) first show that $|z + w|^2 = |z|^2 + |w|^2 + \bar{w}z + \bar{z}w$]

$z = a + ib$, $\bar{z} = a - ib$, $w = x + iy$, $\bar{w} = x - iy$

(i) $|z|^2 = (\sqrt{a^2 + b^2})^2 = a^2 + b^2$

$z\bar{z} = (a + ib)(a - ib) = a^2 - aib + aib + b^2 = a^2 + b^2$

$\therefore |z|^2 = z\bar{z}$

(ii) $|Re(z)| = |a| = \sqrt{a^2}$

$|z| = \sqrt{a^2 + b^2}$ $(b^2 \geqq 0)$

$\therefore |Re(z)| - |z| \geqq 0$

(iii) $z + w = (a + x) + i(b + y)$

$|z| + |w| + 2Re(\bar{z}w)$

$|z + w|^2 = (\sqrt{a^2 + x^2 + b^2 + y^2 + 2ax + 2by})^2 = a^2 + b^2 + x^2 + y^2 + 2(ax + by) \leqq |z|^2 + |w|^2 + |\bar{z}w|^2$

$(|z|^2 + |w|^2 + 2|z||w|)$

$(|z| + |w|)^2 = |z|^2 + |w|^2 + 2|z||w| = a^2 + x^2 + b^2 + y^2 + 2\sqrt{(a^2 + b^2)(x^2 + y^2)}$

$\cancel{\therefore (|z| + |w|)^2 - |z + w|^2 = 2\sqrt{(a^2+b^2)x^2 + (a^2+b^2)y^2} - 2(ax + by)}$

$\cancel{\because (\sqrt{(a^2+b^2)x^2 + (a^2+b^2)y^2})^2 - (ax + by)^2 = (bx + ay)^2 \geqq 0}$

$\therefore |z + w|^2 \leqq (|z| + |w|)^2$

$\therefore |z + w| \leqq |z| + |w|$

Question 10

Find all complex solutions to the following equation

$$1 + z + z^2 + z^3 + z^4 + z^5 = 0$$

$\because z \neq 0$

$\therefore z + z^2 + z^3 + z^4 + z^5 + z^6 = 0$

$\therefore (z + z^2 + z^3 + z^4 + z^5 + z^6) - (1 + z + z^2 + z^3 + z^4 + z^5) = 0$

$\therefore z^6 - 1 = 0$

$\therefore z^6 = 1$

$\because z^6 = r(\cos\theta + i\sin\theta)$

$= r^6(\cos 6\theta + i\sin 6\theta) \quad \therefore r = 1$

$\therefore z_n = \cos\frac{2(n-1)\pi}{6} + i\sin\frac{2(n-1)\pi}{6}$ (n from 1 to 6)

$\therefore z_1 = \cos 0 + i\sin 0 = 1$

$z_2 = \cos\frac{2\pi}{6} + i\sin\frac{2\pi}{6} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$

$z_3 = \cos\frac{4\pi}{6} + i\sin\frac{4\pi}{6} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

$z_4 = \cos\frac{6\pi}{6} + i\sin\frac{6\pi}{6} = -1$

$z_5 = \cos\frac{8\pi}{6} + i\sin\frac{8\pi}{6} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

$z_6 = \cos\frac{10\pi}{6} + i\sin\frac{10\pi}{6} = \frac{1}{2} - i\frac{\sqrt{3}}{2}$

Question 11

Show that z is a solution to the equation

$$3\cosh z + \sinh z = 2$$

if and only if $e^z = \frac{1}{2} \pm \frac{i}{2}$. Hence find all complex solutions to the above equation.

$$3\left(\frac{e^z + e^{-z}}{2}\right) + \left(\frac{e^z - e^{-z}}{2}\right) = 2$$

$$\therefore 2e^z + e^{-z} = 2$$

$$\therefore 2e^z + \frac{1}{e^z} = 2$$

$$\therefore 2(e^z)^2 - 2e^z + 1 = 0$$

$$\therefore e^z = \frac{1}{2} \pm \frac{1}{2}i$$

$$\therefore z = \ln\left(\frac{1}{2} \pm \frac{1}{2}i\right)$$

Question 12

(i) Let $z = 2 + i$. Calculate $\arg(z)$ and $\arg(z-1)$.

(ii) Describe the set

$$\left\{z \in \mathbb{C} : \arg(z-1) = \frac{\pi}{4}\right\}.$$

(iii) Describe the set

$$\left\{z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}\right\}$$

(i) $\arg(z) = \arctan(\frac{1}{2}) + 2n\pi = 0.464 + 2n\pi$, $n = 0, \pm1, \pm2, \pm3 \ldots$

$\arg(z-1) = \arctan(\frac{1}{1}) + 2n\pi = \frac{1}{4}\pi + 2n\pi$, $n = 0, \pm1, \pm2, \pm3 \ldots$

(ii) $\arg(z-1) = \frac{1}{4}\pi$

This set of points is a half-line in the complex plane starting at -1 and extending to infinity in the direction making the angle $\theta = \frac{1}{4}\pi$ with the positive real axis

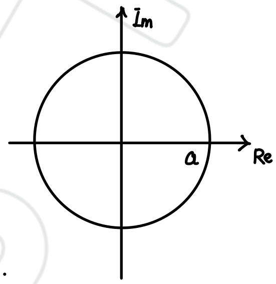
(iii) $\arg(\frac{z-1}{z+1}) = \arg(z-1) - \arg(z+1) = \frac{\pi}{2}$

This set of points is the difference of the angle between two half-line in the complex plane start at -1 and 1 and extending to infinity with the positive real axis

Question 13

The Joukowski transformation in aerofoil theory is

$$J(z) = z + \frac{1}{z}, \qquad z \neq 0.$$



(i) Let C_1 be the circle

$$C_1 = \Big\{ x + iy \in \mathbb{C} : x = a\cos\theta, y = a\sin\theta,\ \theta \in [0, 2\pi) \Big\}.$$

of radius $a > 0$ centre the origin. Find the image of C_1 under the transformation J (i.e. the set of points $J(z)$ such that $z \in C_1$).

(ii) Find the image of the circle

$$C_2 = \Big\{ x + iy \in \mathbb{C} : x = -0.05 + 1.05\cos\theta, y = 0.05 + 1.05\sin\theta,\ \theta \in [0, 2\pi) \Big\}.$$

under the transformation J. Use Matlab (or otherwise) to plot the image $J(C_2)$.

(i) $Z = x + iy \quad \therefore \frac{1}{Z} = \frac{1}{x+iy}$

$\therefore Z + \frac{1}{Z} = x + iy + \frac{1}{x+iy} = x + iy + \frac{x - iy}{x^2+y^2}$

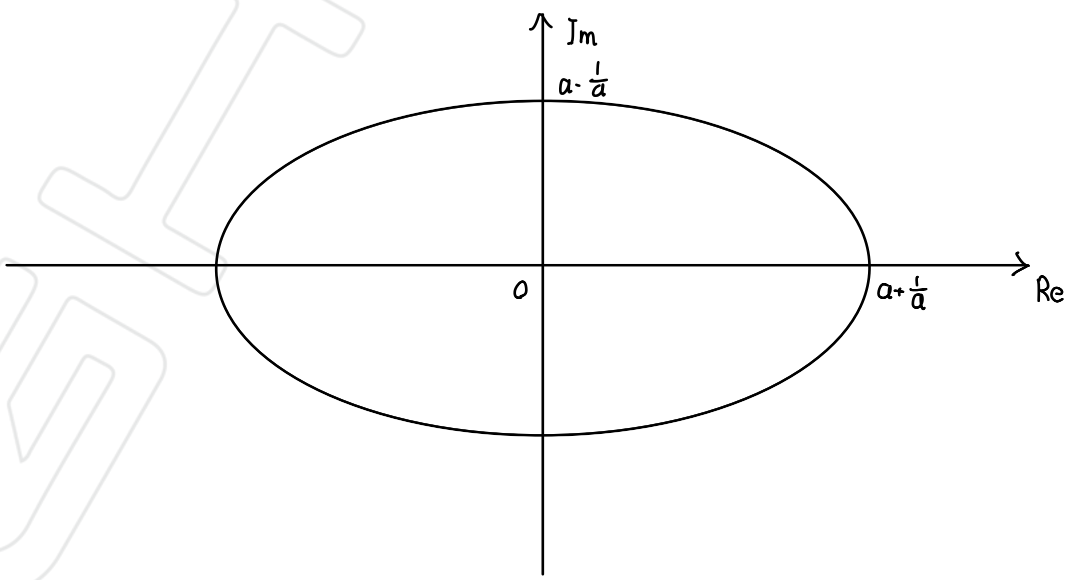
$= (x + \frac{x}{x^2+y^2}) + (y - \frac{y}{x^2+y^2})i$

$\because x = a\cos\theta,\ y = a\sin\theta$

$\therefore Z + \frac{1}{Z} = (a\cos\theta + \frac{a\cos\theta}{a^2}) + (a\sin\theta - \frac{a\sin\theta}{a^2})i$

$= \underbrace{(a + \frac{1}{a})\cos\theta}_{u} + i\underbrace{(a - \frac{1}{a})\sin\theta}_{v} \qquad \therefore \frac{u^2}{(a+\frac{1}{a})} + \frac{v^2}{(a-\frac{1}{a})} = \cos^2\theta + \sin^2\theta = 1$

$J(C_1) = \Big\{ (a+\frac{1}{a})\cos\theta + i(a-\frac{1}{a})\sin\theta \in \mathbb{C},\ \theta \in [0, 2\pi) \Big\}$



(ii) $z = x + iy \quad \therefore \frac{1}{z} = \frac{1}{x+iy}$

$$\therefore J(z) = z + \frac{1}{z} = x + iy + \frac{1}{x+iy}$$

$$= x + iy + \frac{x - iy}{x^2+y^2}$$

$$= \left(x + \frac{x}{x^2+y^2}\right) + i\left(y - \frac{y}{x^2+y^2}\right)$$

$\because x = -0.05 + 1.05\cos\theta$

$y = 0.05 + 1.05\sin\theta$

$$\therefore x^2+y^2 = (0.05)^2 - 2(0.05)(1.05\cos\theta) + (1.05\cos\theta)^2 + (0.05)^2 + 2(0.05)(1.05\sin\theta) + (1.05\sin\theta)^2$$

$$= \frac{1}{200} + \frac{441}{400} + 0.105(\sin\theta - \cos\theta) = 1.1075 + 0.105(\sin\theta - \cos\theta)$$

$$J(C_2) = \left\{ z + \frac{1}{z} \in \mathbb{C},\ \theta \in [0, 2\pi) \right\}$$